Mechtron 3TB4: Embedded Systems Design II
Tutorial 1 for Lab 1

Introduction to Quartus Prime and DE1-SoC

Reports Due: On Avenue, by 11:59PM on Jan 30, 2026 (along with the pre-lab report - ONE PDF)

Goals

- Introduce the Quartus Prime software
- Learn how to describe simple circuitry in Verilog HDL
  Note: **The following documents may help you with your lab. Please go over them at your convenience, in addition to the class notes.**
- Introduction to the software
  Go to Intel's web site for overview of Quartus Prime
- After you install the software (see "Activities" below) you may find more help topics under the "`Help`" menu.
- You may also wish to consult http://www.asic-world.com/verilog/index.html for lessons and references on Verilog HDL.

Lab Equipment and Software

In this tutorial you will be introduced to the new development board that we will use for future labs. Altera's DE1-SoC development board is built around an FPGA device that can be programmed to implement arbitrary logic circuits. The FPGA is connected to many on-board peripherals, as shown in Figure 1.

Peripheral connections

This device contains many peripherals that can be used with the FPGA. The file `DE1-SoC.qsf`, a copy of which can be found in `refs` folder for this lab, provides a pin-map to connect the FPGA's output ports to the surrounding peripherals. For this lab, you will be required to use the DE1-SoC pin assignments to interface with the peripherals of the DE1-SoC development board.

Cyclone V FPGA

The Cyclone V FPGA will be used to implement the hardware logic.

Software Environment

The software environment consists of the Quartus Prime CAD tool.

As a part of your preparation, you will complete a tutorial that will introduce you to the Quartus Prime user interface.

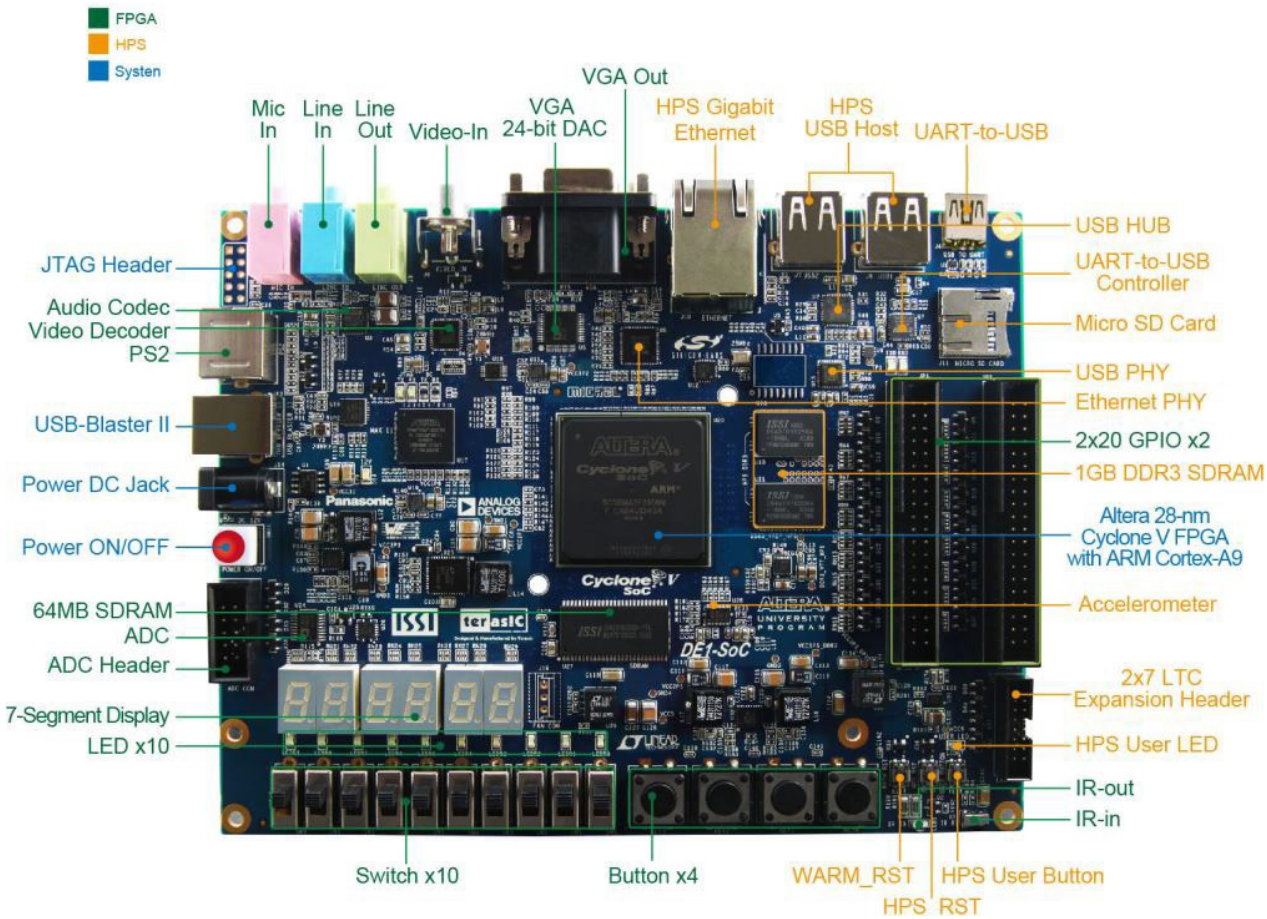

Figure 1: Altera DE1-SoC Board

Activities

Pre-tutorial

The following activities must be completed by each student **independently** before attending this tutorial.

1. Install either the Standard Edition or the Lite Edition of Quartus Prime on your personal computer, available from Intel's website (https://www.intel.com/content/www/us/en/products/details/fpga/development-tools/quartus-prime/resource.html).

   It is recommended to download and install version 23.1, since this is the version that is installed on our department lab computers. Quartus Prime Lite Edition is a free version. If you install the Standard Edition, you need to set up the license. To do so, click `Tools | License Setup...` from Quartus, input "`27000@alteralm.cas.mcmaster.ca`" (without quotation marks) in the "`License File`" box. If you are off campus, you need to use VPN to connect to our department licence server

to use the license file.

For our lab projects, you need to download and install at least these components 1) Quartus Prime. 2) Questa-Intel. 3) Cyclone V device support. More details for installing the software tools, for setting up the software and for setting up the license for using the software for our labs are provided in the `Notes for Installing Software Tools for MT3TB4 Labs (InstallSoftwareTools.pdf)`, which is included in the `refs` folder once you download and unzip the Lab1.zip file.

2. The FAQ for SE2DA4 at http://www.cas.mcmaster.ca/~leduc/FAQ.html contains some useful information about the DE1-SoC board. This FAQ also provides details for setting up the license and some instructions for programming the board and simulating a project. Reading this FAQ should help you with your labs.

3. Complete the tutorial "`Quartus Prime Introduction Using Verilog Designs`" available on the course web page. You do not need to complete Sections 8 and 9 if you do not have the DE1-SoC board at home.

In the Lab

1. Complete the tutorial "`Simulate a Verilog Design Using a Waveform Imput File`" available on the course web page.

   Please Note: In some situations, some nodes may be "synthesized away" during the synthesis and analysis process, as a result it is not possible to observe these nodes during simulation or the signal probing process. Verilog HDL provides some synthesis attributes to direct Analysis & Synthesis to keep them intact. These attributes include "/*synthesis keep */", "/*synthesis preserve */" and "/*synthesis noprune */". For more details about these three synthesis attributes, please read the appropriate Quartus help files or other materials available on the web.

   In case there is no synthesis attribute seems to work to prevent a node to be " synthesized away", please temporarily declare them as output ports. In this lab project, there is no node that will be "synthesized away", but it may happen with your later lab projects.

   Note: Starting from Quartus Prime version 17.0, timing simulations are not supported for the Cyclone V FPGA. For a project that is set up for Cyclone V, the result of running a timing simulation will be identical to the functional simulation.

2. Work in pairs to create a new Quartus project, as you learned in the "`Quartus Prime Introduction Using Verilog Designs`" tutorial.

3. Compile the project and connect the DE1-SoC board to its power supply.

4. Follow Section 7 of the tutorial for pin assignment. **Alternatively**, you can import pin assignments from the `DE1-SoC.qsf` file provided on the course web page. To assign pins by importing the file `DE1-SoC.qsf`, you need to use the DE1-SoC peripheral names as in the file `DE1-SoC.qsf`. In the pin assignment file `DE1-SoC.qsf`, the pins are assigned by sentences like: "`set_location_assignment PIN_AB12 -to SW[0]`". In this sentence, the "**PIN_AB12**" is the pin name, the "**SW[0]**" is the peripheral name. The'**SW[0]**" is the first toggle switch on the DE1-SoC board.

5. **Before compiling, make sure that all unused pins are reserved as "Input tri-stated"**. This option is available under `Assignments | Device` > `Device and Pin Options` > `Unused Pins`.

6. Complete Sections 8 and 9 of the `Quartus Prime Introduction Using Verilog Designs` tutorial.

7. Show the compiled circuit to one of the TAs and take a screenshot of the compilation report for inclusion in your report.

8. Use functional simulation to verify the intended function of the circuit.

9. Show the result of your simulation to one of the TAs and take screenshots for your report.

Report

Describe what you did in this tutorial, and include the screen shots taken during various experiments. You are also required to submit the pre-lab report as described in the Lab 1 document. This material must be submitted at the start of Lab 1.